

## GEMINI Bill of Materials

| ITEM NO. | PART NO.  | DESCRIPTION                                                                                 | REQ. QTY | ORDER QTY | UNIT PRICE | TOTAL PRICE | COMMENTS                                                                             |
|----------|-----------|---------------------------------------------------------------------------------------------|----------|-----------|------------|-------------|--------------------------------------------------------------------------------------|
| 1        | 89015K259 | 6061 Aluminum Sheet 0.160" Thick, 8" x 8"                                                   | 3        | 4         | \$19.49    | \$77.96     | Roof, Upper Outer Skirt                                                              |
| 2        | 89015K255 | 6061 Aluminum Sheet 0.160" Thick, 6" x 6"                                                   | 1        | 1         | \$12.47    | \$12.47     | Inner Skirt                                                                          |
| 3        | 89155K221 | Oversized 6061 Aluminum Sheet 1/4" Thick, 8" x 8"                                           | 1        | 2         | \$17.06    | \$34.12     | All other parts except Deflector Plates                                              |
| 4        | 8975K613  | 6061 Aluminum Bar 5/16" Thick x 1" Wide x 12" Long                                          | 1        | 1         | \$4.18     | \$4.18      | 3.6" required. Deflector Plates. Ordered 8976K856 (same)                             |
| 5        | 92196A103 | 4-40, 1/8" Long, 18-8 SS Socket Head Screw. Pack of 100                                     | 14       | 1         | \$6.10     | \$6.10      | Shield mounting and Deflector Plate power.                                           |
| 6        | 90550A122 | M4 x 0.7 mm, 10 mm Long, 18-8 SS Vented Flat Head Screw. Pack of 5                          | 4        | 1         | \$13.10    | \$13.10     | Inner Skirt mounting to Rotational Stage                                             |
| 7        | 93330A541 | 4-40, 3" Long, Aluminum Female-Female Round Standoff                                        | 4        | 6         | \$2.17     | \$13.02     | Roof Mounting to Upper Outer Skirt. Ordered 4 on 12/8/25                             |
| 8        | 94335A115 | 6-32, 1/2" Long, Glazed L5 Ceramic Female-Female Round Standoff                             | 4        | 6         | \$5.65     | \$33.90     | Deflector Plate Mounting Standoffs. Ordered 4 on 12/8/25                             |
| 9*       | 7113K148  | 22-18 Wire Gauge, #4 Screw Size, Tin-Plated Copper Ring Terminals Noninsulated. Pack of 100 | 2        | 1         | \$16.02    | \$16.02     | Power delivery to Deflector Plates.                                                  |
| 10*      | 92210A077 | 2-56, 1/4" Long, 18-8 SS Flat Head Screw 82 Degree Countersink. Pack of 100                 | 4        | 1         | \$5.83     | \$5.83      | For mounting Aperture Plates                                                         |
| 11*      | 92210A105 | 4-40, 1/4" Long, 18-8 SS Flat Head Screw 82 Degree Countersink. Pack of 100                 | 18       | 1         | \$4.43     | \$4.43      | Workhorse. Standard fastener for most of GEMINI. Ordered anyways (unsure of lab qty) |
| 12*      | 93235A144 | Vented Socket Head Screw 6-32 Thread Size, 1/4" Long. Pack of 5                             | 8        | 2         | \$5.97     | \$11.94     | For Deflector Plate ceramic standoffs                                                |
| 13       | 92510A460 | #8, 1/8" Long Aluminum Unthreaded Spacer                                                    | 4        | 8         | \$0.38     | \$3.04      | For raising Inner Skirt                                                              |
| 14       | 8974K105  | 6061 Aluminum Rod 3/16" Diameter x 1/2 Foot Long                                            | 1        | 1         | \$1.60     | \$1.60      | 0.687" each (x4). Dummy hardware for when motor is detached. Ordered 8974K21 (same)  |
| 15       | 92125A138 | M3 x 0.5 mm, 25 mm Long, 18-8 SS Flat Head Screw. Pack of 100                               | 4        | 1         | \$11.38    | \$11.38     | Require 4. Dummy hardware for when motor is detached                                 |
| 16       | 8803A383  | 0.1985" Reamer, High-Speed Steel                                                            | 1        | 1         | \$24.52    | \$24.52     | For creating Aperture Rotation Plate collar hole                                     |
| 17       | 93033A107 | M3 x 0.5 mm 18-8 SS Flange Nut. Pack of 100                                                 | 4        | 1         | \$7.33     | \$7.33      | For mounting D42.1 Motor                                                             |
| 18       | 91979A207 | 2-56, 1/8" Long, 18-8 SS Set Screw. Pack of 10                                              | 1        | 1         | \$6.79     | \$6.79      | For Aperture Rotation Plate                                                          |
| 19       | 92510A435 | #12, 2-1/4" Long Aluminum Unthreaded Spacer                                                 | 5        | 5         | \$7.73     | \$38.65     | For mounting Upper Outer Skirt to Rotational Stage                                   |
| 20       | 91292A322 | M5 x 0.8 mm, 110 mm Long, 18-8 SS Socket Head Screw. Pack of 10                             | 5        | 1         | \$11.03    | \$11.03     | For mounting Upper Outer Skirt to Rotational Stage                                   |
| 21       | 88805A27  | 1/4" Ball End Mill, Uncoated Carbide, 4 Flute, 2-1/2" Overall Length                        | 1        | 1         | \$20.85    | \$20.85     | For machining fillets                                                                |
| 22       | 2739A38   | 90 Degree Fluteless Smooth-Finish Countersink                                               | 1        | 1         | \$15.40    | \$15.40     | For Inner Skirt M4 countersinks                                                      |
| 23       | 2739A33   | 82 Degree Fluteless Smooth-Finish Countersink                                               | 1        | 1         | \$15.40    | \$15.40     | For all other countersinks                                                           |

**TOTAL: \$389.06**

\* indicates part already exists in the Space Physics Lab

## GEMINI Bill of Materials (Continued)

| OTHER COMPONENTS (INFRASTRUCTURE & NON-CRITICAL) |          |                                                                |          |           |            |                |                                                    |
|--------------------------------------------------|----------|----------------------------------------------------------------|----------|-----------|------------|----------------|----------------------------------------------------|
| ITEM NO.                                         | PART NO. | DESCRIPTION                                                    | REQ. QTY | ORDER QTY | UNIT PRICE | TOTAL PRICE    | COMMENTS                                           |
| 23                                               | D42.1-H  | AML UHV Stepper Motor. Dry, lubricated hybrid ceramic bearings | 1        | 1         | \$1,080    | \$1,080        | For rotating aperture plates                       |
| 24                                               | SMD3     | AML Single-Axis Bipolar Stepper Motor Drive                    | 1        | 1         | \$840      | \$840          | For controlling D42.1-H motor                      |
| 25*                                              | 3394A    | Quantar 2-Layer 40 mm MCP                                      | 1        | 0         | N/A        | N/A            |                                                    |
| 26*                                              |          | MCP #0 Mounting Screws                                         | 4        | 0         | N/A        | N/A            | For mounting MCP                                   |
| 27*                                              |          | RV120 Rotational Stage                                         | 1        | 0         | N/A        | N/A            |                                                    |
| 28*                                              |          | CoDICE 0.9895" Spacer                                          | 1        | 0         | N/A        | N/A            |                                                    |
| 29*                                              |          | 0.010" Thick Aluminum Shim Stock                               | 1        | 0         | N/A        | N/A            | For Shield and CoDICE Spacer Shim                  |
| 30*                                              |          | SHV5 Connector                                                 | 2        | 0         | N/A        | N/A            |                                                    |
| 31*                                              |          | High Voltage insulated wiring                                  |          | 0         | N/A        | N/A            |                                                    |
| 32*                                              |          | High Voltage Power Supply                                      | 2        | 0         | N/A        | N/A            | For Deflector Plates                               |
| 33*                                              |          | Low Voltage Power Supply                                       | 1        | 0         | N/A        | N/A            | For Deflector Plates measuring secondary electrons |
| <b>TOTAL:</b>                                    |          |                                                                |          |           |            | <b>\$1,920</b> |                                                    |

\* indicates part already exists in the Space Physics Lab